- Solvent of higher dielectric constant show high conductance in solution.
- Conductance is inversely proportional to the Viscosity of the medium. i.e., conductivity increases with the decrease in viscosity.
- If the temperature of the electrolytic solution increases, conductance also increases. Increase in temperature increases the kinetic energy of the ions and decreases the attractive force between the oppositely charged ions and hence conductivity increases.
- Molar conductance of a solution increases with increase in dilution. This is because, for a strong electrolyte, interionic forces of attraction decrease with dilution. For a weak electrolyte, degree of dissociation increases with dilution.

9.1.4 Measurement of conductivity of ionic solutions

We have already learnt to measure the specific resistance of a metallic wire using a metre bridge in your physics practical experiment. We know that it works on the principle of wheatstone bridge. Similarly, the conductivity of an electrolytic solution is determined by using a wheatstone bridge arrangement in which one resistance is replaced by a conductivity cell filled with the electrolytic solution of unknown conductivity.

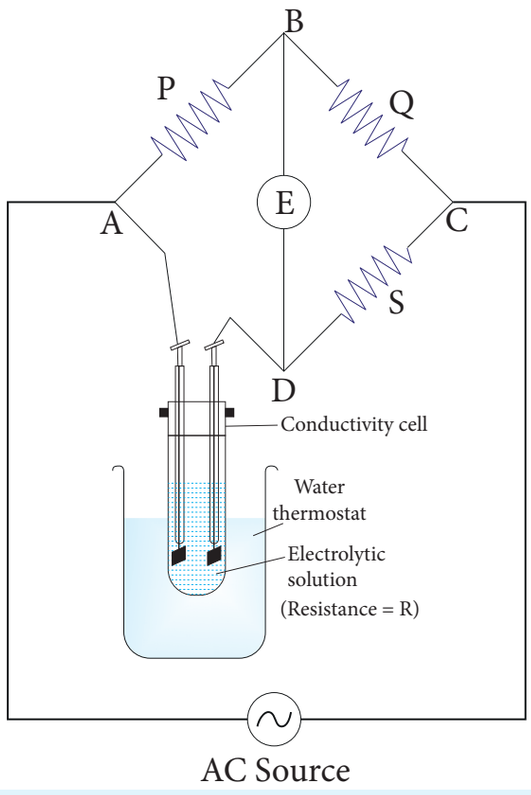


Fig 9.3 Schematic diagram of a conductivity cell in a wheatstone bridge circuit

In the measurement of specific resistance of a metallic wire, a DC power supply is used. Here, if we apply DC current through the conductivity cell, it will lead to the electrolysis of the solution taken in the cell. So, AC current is used for this measurement to prevent electrolysis.

A wheatstone bridge is constituted using known resistances P, Q, a variable resistance S and conductivity cell (Let the resistance of the electrolytic solution taken in it be R) as shown in the figure 9.3. An AC source (550 Hz to 5 KHz) is connected between the junctions A and C. Connect a suitable detector E (Such as the telephone ear piece detector) between the junctions 'B' and 'D'.

The variable resistance 'S' is adjusted until the bridge is balanced and in this conditions there is no current flow through the detector.

Under balanced condition,

$$\frac{P}{Q} = \frac{R}{S}$$

$$\therefore R = \frac{P}{Q} \times S \qquad \text{.....(9.10)}$$

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